

IMAT MOCK EXAM 1 — "Baseline"

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 47/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "Vaccination operates on the principle of immunological memory. Upon first exposure to an antigen, naïve lymphocytes proliferate and differentiate, generating both effector cells that clear the pathogen and long-lived memory cells. A subsequent encounter with the same antigen triggers a faster and stronger response because memory cells respond at lower antigen concentrations and divide more rapidly than naïve cells."

Which of the following is not supported by the passage?

- A) The secondary immune response exceeds the primary one in magnitude.
- B) Effector cells contribute to pathogen clearance.
- C) Naïve lymphocytes proliferate during the primary response.
- D) Memory cells persist after the initial infection has been cleared.
- E) Memory cells require a higher antigen concentration to activate than naïve cells.

2. "Enzymes lower the activation energy of a reaction without altering the free energy difference between substrates and products. Consequently, an enzyme changes the rate at which equilibrium is reached but never the position of equilibrium."

Which statement is a valid inference?

- A) The ratio of products to substrates at equilibrium is unchanged by the enzyme.
- B) Adding an enzyme increases the equilibrium constant.
- C) Enzymes shift the equilibrium toward products.
- D) A reaction that is unfavourable thermodynamically can be made favourable by adding more enzyme.
- E) An enzyme can make an endergonic reaction exergonic.

3. (General knowledge) Which of the following Italian scientists is best known for discovering a method of staining nervous tissue that allowed the visualisation of individual neurons, a contribution that shared the 1906 Nobel Prize in Physiology or Medicine?

- A) Antonio Meucci
- B) Rita Levi-Montalcini
- C) Galileo Galilei
- D) Camillo Golgi
- E) Guglielmo Marconi

4. Five friends — A, B, C, D, E — sit in a row. B sits immediately to the right of A. C is not adjacent to A. D sits at one of the ends. E sits to the left of D. Which of the following must be true?

- A) E is not at an end.
- B) D sits at the far right.
- C) B is to the left of C.
- D) A is at the far left.
- E) C sits between B and E.

5. All patients with disease X have elevated marker M. Some patients with disease Y also have elevated marker M. No patient with disease X has disease Y. From these, which conclusion necessarily follows?

- A) No patient with disease Y has elevated M.
- B) Every patient with elevated M has disease X or Y.
- C) Some patients with disease Y do not have disease X.
- D) Anyone with elevated M has disease X.
- E) Patients with disease X always have disease Y.

6. In a certain code, READING is written as SFB EJ OH. How is MONKEY written?
A) NPO LEZ B) NPO MFZ C) NPO LFY D) NPO LFZ E) NQO LFZ
7. A clock shows 3:15. What is the angle between the hour and minute hands?
A) 0° B) 7.5° C) 15° D) 22.5° E) 30°
8. If the day after tomorrow is Wednesday, what day was it three days before yesterday?
A) Saturday B) Tuesday C) Friday D) Sunday E) Thursday
9. A sequence begins 2, 6, 12, 20, 30, ... What is the next term?
A) 36 B) 40 C) 42 D) 44 E) 50

Section B - Biology (Q10-32)

10. In which cellular compartment does the Krebs (citric acid) cycle take place?
**A) Mitochondrial matrix
B) Rough endoplasmic reticulum
C) Mitochondrial intermembrane space
D) Nucleus
E) Cytosol**
11. How many net ATP molecules are produced by substrate-level phosphorylation during glycolysis per molecule of glucose?
A) 0 B) 1 C) 2 D) 4 E) 6
12. Which molecule serves as the final electron acceptor of the mitochondrial electron transport chain?
A) Water B) Pyruvate C) FAD D) Oxygen E) NAD⁺
13. The ATP synthase enzyme is located in the:
**A) Inner mitochondrial membrane
B) Cytosol
C) Outer mitochondrial membrane
D) Thylakoid stroma
E) Nuclear envelope**
14. During which phase of cellular respiration is the majority of CO₂ released?
**A) Oxidative phosphorylation
B) Pyruvate oxidation and the Krebs cycle
C) Lactic acid fermentation
D) Glycolysis
E) Substrate-level phosphorylation**
15. Which organelle is the primary site of protein synthesis for proteins destined for secretion?
**A) Peroxisome
B) Nucleolus
C) Smooth endoplasmic reticulum
D) Lysosome
E) Rough endoplasmic reticulum**
16. Which of the following is a function of the Golgi apparatus?
**A) Storage of genetic information
B) Detoxification of hydrogen peroxide
C) ATP production via oxidative phosphorylation
D) Synthesis of ribosomal RNA
E) Modification, sorting, and packaging of proteins**

- 17.** Which structure is responsible for the selective permeability of the cell membrane?
- A)** The cytoskeleton
 - B)** The nuclear pore complex alone
 - C)** The cell wall of cellulose
 - D)** The phospholipid bilayer with embedded proteins
 - E)** The peptidoglycan layer
- 18.** Which transport mechanism requires the direct hydrolysis of ATP to move a substance against its concentration gradient?
- A)** Simple diffusion
 - B)** Facilitated diffusion via channels
 - C)** Primary active transport
 - D)** Passive carrier-mediated diffusion
 - E)** Osmosis
- 19.** Which of the following correctly describes the induced-fit model of enzyme action?
- A)** The substrate is permanently altered and the enzyme consumed.
 - B)** The enzyme changes shape upon substrate binding to improve the fit.
 - C)** The enzyme raises ΔG of the reaction.
 - D)** Competitively bound inhibitors bind allosteric sites.
 - E)** The active site is rigid and only one substrate can bind.
- 20.** A competitive inhibitor of an enzyme:
- A)** Increases K_m but does not change V_{max} .
 - B)** Decreases V_{max} but does not change K_m .
 - C)** Binds covalently to the active site, inactivating the enzyme permanently.
 - D)** Cannot be overcome by increasing substrate concentration.
 - E)** Binds reversibly at a site other than the active site.
- 21.** Which type of muscle is involuntary, striated, and has branched cells connected by intercalated discs?
- A)** Myoepithelium
 - B)** Smooth muscle
 - C)** Cardiac muscle
 - D)** Skeletal muscle
 - E)** Tendons
- 22.** The functional unit of contraction in skeletal muscle is the:
- A)** Sarcolemma
 - B)** Sarcoplasm
 - C)** Sarcoplasmic reticulum
 - D)** Myofibril
 - E)** Sarcomere
- 23.** During the sliding filament mechanism, myosin binds to:
- A)** Myosin-binding site on titin
 - B)** Actin
 - C)** Troponin
 - D)** Tropomyosin
 - E)** Calcium directly
- 24.** Oxygen is transported in blood primarily:
- A)** Within leucocytes
 - B)** Bound to albumin
 - C)** Dissolved in plasma
 - D)** Bound to haemoglobin in red blood cells
 - E)** As bicarbonate ions

- 25.** In which part of the human digestive system does the majority of nutrient absorption occur?
A) Small intestine
B) Large intestine
C) Stomach
D) Oesophagus
E) Liver
- 26.** Which hormone, produced by the beta cells of the pancreas, lowers blood glucose concentration?
A) Glucagon **B)** Adrenaline **C)** Thyroxine **D)** Insulin **E)** Cortisol
- 27.** Which hormone is released from the posterior pituitary and causes water reabsorption in the kidney collecting duct?
A) Antidiuretic hormone (ADH / vasopressin)
B) Atrial natriuretic peptide
C) Growth hormone
D) Aldosterone
E) Prolactin
- 28.** Which bone of the upper limb is the humerus? It is the:
A) Thigh bone
B) Upper arm bone
C) Shin bone
D) Forearm bone (thumb side)
E) Collar bone
- 29.** In DNA replication, the enzyme that unwinds the double helix at the replication fork is:
A) Telomerase **B)** Ligase **C)** DNA polymerase **D)** Primase **E)** Helicase
- 30.** In a diploid organism with $2n = 46$, how many chromosomes are present in a mature gamete?
A) 23 **B)** 44 **C)** 46 **D)** 48 **E)** 92
- 31.** A man with blood group AB and a woman with blood group O have a child. Which blood groups are possible in their offspring?
A) A and AB
B) AB only
C) A and B only
D) O only
E) A, B, AB, and O
- 32.** During translation, the codon AUG serves to:
A) Bind release factors
B) Code for tryptophan only
C) Act as the start codon and code for methionine
D) Recruit RNA polymerase
E) Terminate translation

Section C - Chemistry (Q33-47)

- 33.** What is the oxidation number of sulfur in H_2SO_4 ?
A) -2 **B)** 0 **C)** +2 **D)** +4 **E)** +6
- 34.** In the reaction $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, which species is reduced?
A) Zn **B)** Cu^{2+} **C)** Zn^{2+} **D)** Cu (metal) **E)** SO_4^{2-}
- 35.** An acidic aqueous solution at 25°C has a pH of 3. The concentration of H^+ is:
A) $1 \times 10^{-3} \text{ M}$ **B)** $1 \times 10^3 \text{ M}$ **C)** $1 \times 10^{-11} \text{ M}$ **D)** $3 \times 10^{-1} \text{ M}$ **E)** 3 M

36. What is the pH of a 0.01 M solution of a strong monoprotic acid (e.g., HCl) at 25 °C?
A) 1 B) 2 C) 3 D) 12 E) 13
37. Which type of bond involves the complete transfer of electrons from one atom to another?
**A) Metallic bond
B) Van der Waals interaction
C) Hydrogen bond
D) Ionic bond
E) Covalent bond**
38. How many electrons are shared in a double covalent bond between two atoms?
A) 1 B) 2 C) 3 D) 4 E) 6
39. At constant temperature and pressure, the volume of a gas is directly proportional to the number of moles. This is:
A) Boyle's law B) Dalton's law C) Avogadro's law D) Charles's law E) Henry's law
40. Calculate the number of molecules in 0.5 mol of N₂ (Avogadro's number = 6.022×10^{23}).
A) 1.20×10^{24} B) 1.50×10^{23} C) 3.01×10^{23} D) 6.02×10^{23} E) 6.02×10^{22}
41. Which element is in Group 17 and Period 3 of the periodic table?
A) Fluorine B) Iodine C) Sulfur D) Bromine E) Chlorine
42. Which subatomic particle has no electric charge and a mass of approximately 1 amu?
A) Neutron B) Proton C) Photon D) Positron E) Electron
43. For the equilibrium $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H < 0$. Increasing the pressure at constant temperature will:
**A) Shift equilibrium to the left (fewer moles side is reactants — wrong)
B) Shift equilibrium to the right, producing more NH₃
C) Decrease the rate only
D) Increase the temperature
E) Have no effect**
44. The functional group -COOH characterises which class of organic compounds?
**A) Alcohols
B) Ethers
C) Aldehydes
D) Carboxylic acids
E) Ketones**
45. Which of the following compounds is an alkane?
A) C₂H₄ B) C₂H₂ C) C₆H₆ D) C₃H₆ E) C₄H₁₀
46. The rate of a chemical reaction generally increases with temperature because:
**A) The activation energy decreases.
B) The reaction becomes exothermic.
C) The equilibrium constant always increases.
D) Catalysts form spontaneously.
E) More collisions occur with energy $\geq E_a$ per unit time.**
47. What is the molar concentration of a solution containing 0.2 mol of NaCl dissolved to make 500 mL of solution?
A) 0.1 M B) 0.2 M C) 0.4 M D) 0.5 M E) 1.0 M

Section D - Mathematics (Q48-54)

48. Solve for x : $\sqrt{2x - 1} = 3$.

- A) $x = 2$ B) $x = 3$ C) $x = 6$ D) $x = 5$ E) $x = 4$

49. Solve: $2x - 5 < 3x + 1$.

- A) $x > -6$ B) $x < 6$ C) $x < -6$ D) $x = -6$ E) $x > 6$

50. In a triangle, two angles are 50° and 60° . What is the third angle?

- A) 50° B) 60° C) 70° D) 80° E) 90°

51. If $f(x) = 2x^2 - 3x + 1$, what is $f(2)$?

- A) 1 B) 2 C) 3 D) 5 E) 7

52. A circle has radius 5 cm. What is its area? (Take $\pi = 3.14$.)

- A) 15.7 cm^2 B) 25 cm^2 C) 31.4 cm^2 D) 62.8 cm^2 E) 78.5 cm^2

53. A bag contains 3 red and 2 blue balls. One ball is drawn at random. What is the probability it is blue?

- A) $1/5$ B) $1/2$ C) $2/5$ D) $2/3$ E) $3/5$

54. A rectangle has length 12 cm and diagonal 13 cm. What is its perimeter?

- A) 24 cm B) 28 cm C) 30 cm D) 34 cm E) 36 cm

Section E - Physics (Q55-60)

55. A car travels 120 km in 2 hours at constant velocity. What is its speed in m/s?

- A) 16.7 m/s B) 33.3 m/s C) 50 m/s D) 60 m/s E) 120 m/s

56. Three resistors of 2Ω , 3Ω , and 6Ω are connected in parallel. What is the equivalent resistance?

- A) 0.5Ω B) 1Ω C) 3.66Ω D) 6Ω E) 11Ω

57. A current of 2 A flows through a 5Ω resistor. What is the power dissipated?

- A) 5 W B) 10 W C) 20 W D) 50 W E) 100 W

58. A book rests on a table. The reaction force of the table on the book is an example of which law?

- A) Hooke's law
B) Newton's third law
C) Newton's second law only
D) Newton's first law only
E) Ohm's law

59. The gravitational force between two bodies is $F = Gm_1m_2/r^2$. If the distance between the two bodies is halved, the gravitational force becomes:

- A) $2F$ B) F C) $F/2$ D) $4F$ E) $F/4$

60. The pressure at depth h in a fluid of density ρ is given by $P = \rho gh$. If $\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$, and $h = 10 \text{ m}$, what is the gauge pressure?

- A) 9,800 Pa B) 98 Pa C) 98,000 Pa D) 980 Pa E) 980,000 Pa

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	E	2	A	3	D	4	B	5	C
6	D	7	B	8	E	9	C	10	A
11	C	12	D	13	A	14	B	15	E
16	E	17	D	18	C	19	B	20	A
21	C	22	E	23	B	24	D	25	A
26	D	27	A	28	B	29	E	30	A
31	C	32	C	33	E	34	B	35	A
36	B	37	D	38	D	39	C	40	C
41	E	42	A	43	B	44	D	45	E
46	E	47	C	48	D	49	A	50	C
51	C	52	E	53	C	54	D	55	A
56	B	57	C	58	B	59	D	60	C